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# 1 Install Required Libraries
!pip install --quiet --upgrade google-genai transformers

# 2 Imports
import os
from google import genai
from transformers import pipeline

# 3 Secure API Key Input
os.environ["GOOGLE_API_KEY"] = input("Enter your Google API Key: ")

# 4 Initialize Gemini Client
gemini_client = genai.Client(api_key=os.getenv("GOOGLE_API_KEY"))

# 5 Load HuggingFace GPT-2
print("\nLoading HuggingFace GPT-2 model...")
hf_generator = pipeline("text-generation", model="gpt2")

# 6 Define Model Functions
def query_gemini(prompt):
    response = gemini_client.models.generate_content(
        model="gemini-2.0-flash", # 7 FIXED MODEL NAME
        contents=prompt
    )
    return response.text

def query_huggingface(prompt):
    response = hf_generator(
        prompt,
        max_length=60,
        num_return_sequences=1
    )
    return response[0]["generated_text"]

# =====
# 7 Common Prompt
# =====

prompt = "Write a haiku about AI and nature."

print("\nGenerating responses...\n")

# 8 Generate Outputs
gemini_output = query_gemini(prompt)
hf_output = query_huggingface(prompt)

# 9 Display Outputs
print("## 🎨 Model Outputs\n")

print("### 💡 Gemini Response:\n")
print(gemini_output)

print("\n### 💡 HuggingFace Response:\n")
print(hf_output)

# =====
# 10 Evaluation (Using Gemini as Judge)
# =====

evaluation_prompt = f"""
You are an evaluator. Rank the following responses from best to worst
based on relevance, creativity, and poetic quality.

Prompt: {prompt}

Gemini:
{gemini_output}

HuggingFace:
{hf_output}

Return only ranking in format:

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1st: modelName
2nd: modelName
"""

evaluation = query_gemini(evaluation_prompt)

print("\n## 🏆 Evaluation Result (Best → Worst)\n")
print(evaluation)

# =====
# ✅ END
# =====

```

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!pip install --quiet transformers

from transformers import pipeline
import random

print("Loading GPT-2 model...")
generator = pipeline("text-generation", model="gpt2")

prompt = "Write a haiku about AI and nature."

print("\nGenerating responses...\n")

# Simulate two different models using different temperatures
model_A = generator(prompt, max_length=60, temperature=0.6)[0]["generated_text"]
model_B = generator(prompt, max_length=60, temperature=1.2)[0]["generated_text"]

print("### 💎 Model A (Lower Creativity):\n")
print(model_A)

print("\n### 💎 Model B (Higher Creativity):\n")
print(model_B)

# Simple evaluation logic
print("\n 🏆 Evaluation Result:\n")

if len(model_A.split()) < len(model_B.split()):
    print("1st: Model B")
    print("2nd: Model A")
else:
    print("1st: Model A")
    print("2nd: Model B")

```

